Matematica Terza

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1 Equazioni e disequazioni irrazionali

1.1 Le equazioni irrazionali

Definizione 1 (Equazioni e disequazioni irrazionali)**.** Un'equazione o una disequazione di chiama **irrazionale** quando contiene almeno un radicale nel cui radicando compare l'incognita.

$$\sqrt[n]{A(x)} = B(x)$$

Se n pari

$$\rightarrow \begin{cases} (A(x) \geq 0) & \text{cond. di esistenza} \\ B(x) \geq 0 & \text{concordanza dei segni} \\ A(x) = [B(x)]^n \end{cases}$$

n dispari

$$\sqrt[n]{A(x)} = B(x)$$
$$A(x) = [B(x)]^n$$

Esercizio 1.

$$\sqrt{x^2 - 4x + 3} = 2x - 6$$

$$\begin{cases} x^2 - 4x + 3 \geq 0 & \text{C.E.} \\ 2x - 6 \geq 0 & \text{C.S.} \\ x^2 - 4x + 3 = (2x - 6)^2 \end{cases}$$

$$\begin{cases} (x-3)(x-1) \geq 0 \\ x \geq 3 \\ x^2 - 4x + 3 = 4x^2 - 24x + 36 \end{cases}$$

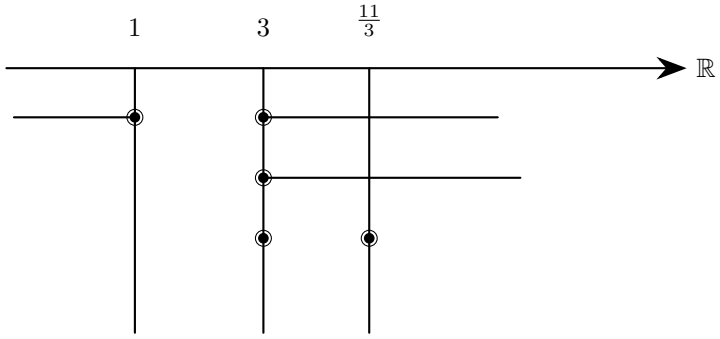
$$\begin{cases} x \leq 1 \vee x \geq 3 \\ x \geq 3 \\ 3x^2 - 20x + 33 = 0 \quad \text{III} \end{cases}$$

$$\text{III} \quad 3x^2 - 20x + 33 = 0$$

$$\frac{\Delta}{4} = \left(\frac{b}{2}\right)^2 - ac = 100 - 99 = 1$$

$$x_{1/2} = \frac{-\frac{b}{2} \pm \sqrt{\Delta}}{a} = \frac{10 \pm 1}{3} = \frac{11}{3} \vee 3$$

$$\begin{cases} x \leq 1 \vee x \geq 3 \\ x \geq 3 \\ x = 3 \vee x = \frac{11}{3} \end{cases}$$



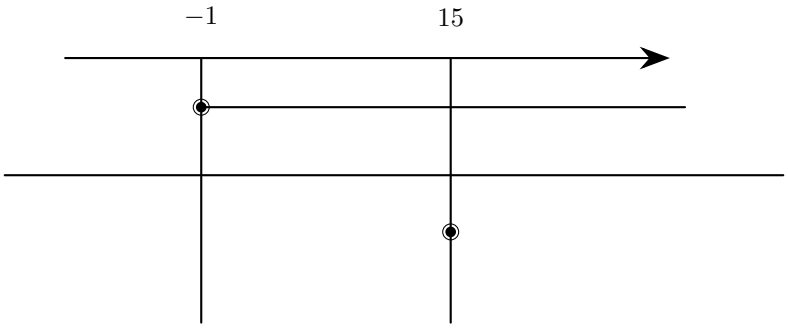
$$\boxed{x = 3 \vee x = \frac{11}{3}}$$

Esercizio 2.

$$\sqrt{x+1} = 4$$

$$\begin{cases} x + 1 \geq 0 & \text{C.E.} \\ 4 \geq 0 & \text{C.S.} \\ x + 1 = 16 \end{cases}$$

$$\begin{cases} x \geq 1 \\ \forall x \in \mathbb{R} \\ x = 15 \end{cases}$$



$$\boxed{x = 15}$$

Esercizio 3.

$$\sqrt{x} - \sqrt{x-5} = -\sqrt{x-3}$$

$$\text{CE: } \begin{cases} x \geq 0 \\ x+5 \geq 0 \\ x-3 \geq 0 \end{cases} \quad \begin{cases} x \geq 0 \\ x \geq -5 \\ x \geq 3 \end{cases}$$

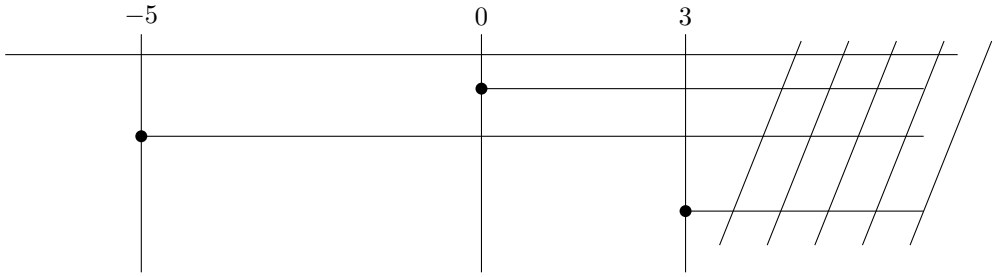


Figura 1: Rappresentazione grafica.

$$\boxed{x \geq 3}$$

$$\sqrt{x} + \sqrt{x-3} = \sqrt{x+5}$$

$$\left(\sqrt{x} + \sqrt{x-3}\right)^2 = \left(\sqrt{x+5}\right)^2$$

$$x + 2\sqrt{x} \cdot \sqrt{x-3} + \not{x} - 3 = \not{x} + 5$$

$$2\sqrt{x\,(x-3)} = -x + 8$$

$$\sqrt{x\,(x-3)} = \frac{-x+8}{2}$$

$$\begin{cases} [x\,(x-3) \geq 0] \\ \frac{-x+8}{2} \geq 0 \\ x\,(x-3) = \left(\frac{-x+8}{2}\right)^2 \end{cases}$$

$$\begin{cases} x \leq 0 \vee x \geq 3 \\ x \leq 8 \\ x^2 - 3x = \frac{x^2 - 16x + 64}{4} \end{cases}$$

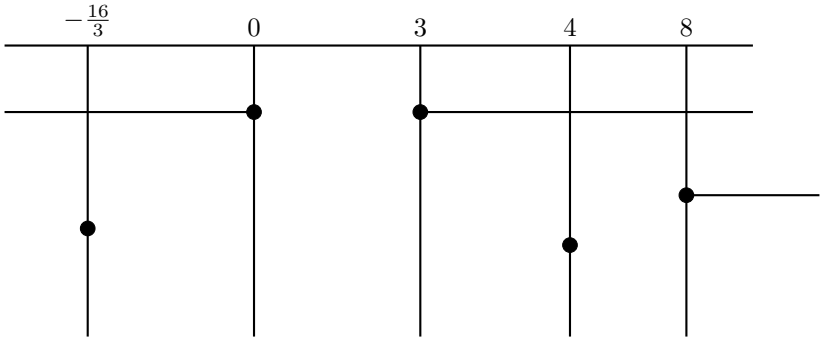
$$\begin{cases} // \\ // \\ 4x^2 - 12x = x^2 - 16x + 64 \end{cases}$$

$$\begin{cases} // \\ // \\ 3x^2 + 4x - 64 = 0 \end{cases}$$

$$\frac{\Delta}{4} = 4 + 192 = 196$$

$$x_{1/2} = \frac{-2 \pm \sqrt{196}}{3} = \frac{-2 \pm 14}{3} = 4 \vee -\frac{16}{3}$$

$$\begin{cases} x \leq 0 \vee x \geq 3 \\ x \leq 8 \\ x = 4 \vee x = -\frac{16}{3} \end{cases}$$



$$\begin{cases} x \geq 3 \\ \boxed{x = 4} \vee \underbrace{x = -\frac{16}{3}}_{\text{n. acc.}} \end{cases}$$

1.2 Disequazioni irrazionali

$$\sqrt{A(x)} < B(x)$$

$$\begin{cases} A(x) \geq 0 \\ B(x) > 0 \\ A(x) < B^2(x) \end{cases}$$

Esercizio 4.

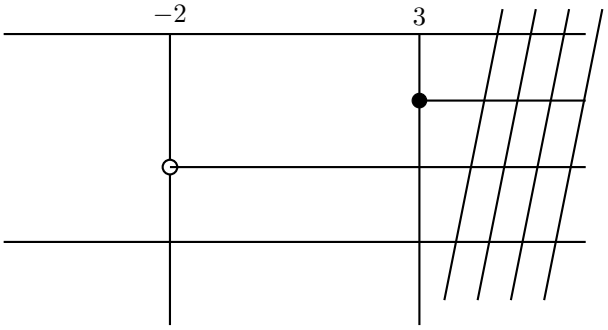
$$\sqrt{x-3} < x+2$$

$$\begin{cases} x-3 \geq 0 \\ x+2 > 0 \\ x-3 < (x+2)^2 \end{cases} \qquad \begin{cases} x \geq 3 \\ x > -2 \\ x-3 < x^2+4x+4 \end{cases}$$

$$\begin{cases} // \\ // \\ x^2+3x+7 > 0 \end{cases}$$

$$\Delta = 9 - 4 \cdot 7 < 0 \quad \underbrace{\text{C.S.}}\text{D.M.}$$

$$\begin{cases} x \geq 3 \\ x > -2 \\ \forall x \in \mathbb{R} \end{cases}$$



$$\boxed{x \geq 3}$$

$$\sqrt{A(x)} < B(x) \qquad \begin{cases} A(x) \geq 0 \\ B(x) > 0 \\ A(x) < B^2(x) \end{cases}$$

$$\sqrt{A(x)} > B(x)$$

$$\begin{cases} B(x) < 0 \\ A(x) \geq 0 \end{cases} \cup \begin{cases} B(x) \geq 0 \\ A(x) > B^2(x) \end{cases}$$

Esercizio 5.

$$\sqrt{9-x^2} > x+3$$

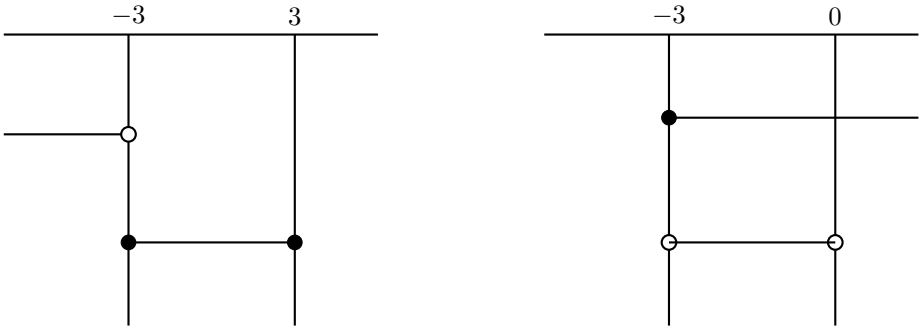
$$\begin{cases} x+3 < 0 \\ 9-x^2 \geq 0 \end{cases} \cup \begin{cases} x+3 \geq 0 \\ 9-x^2 > (x+3)^2 \end{cases}$$

$$\begin{cases} x < -3 \\ -3 \leq x \geq 3 \end{cases} \cup \begin{cases} x \geq -3 \\ \not{9} - x^2 > x^2 + 6x + \not{9} \end{cases}$$

$$\begin{cases} // \\ // \end{cases} \cup \begin{cases} x \geq -3 \\ 2x^2 + 6x < 0 \end{cases}$$

$$\begin{cases} // \\ 2x\,(x+3) < 0 \end{cases}$$

$$\begin{cases} x < -3 \\ -3 \leq x \leq 3 \end{cases} \cup \begin{cases} x \geq -3 \\ -3 < x < 0 \end{cases}$$



$$\nexists x \in \mathbb{R} \cup -3 < x < 0$$

$$-3 < x < 0$$

Esercizio 6.

$$\sqrt{x+3} > x - 5$$

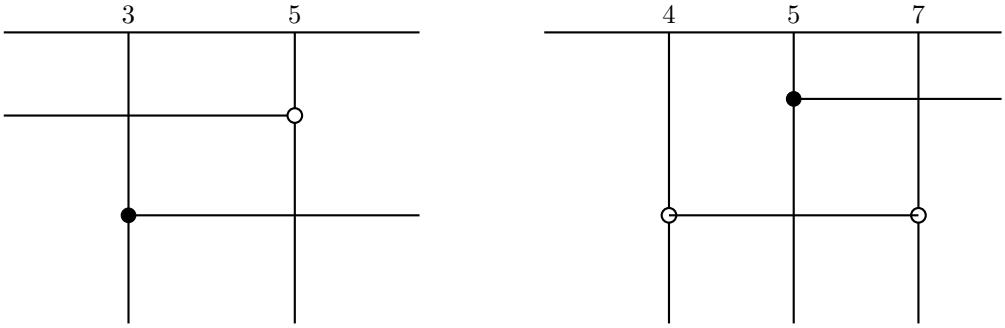
$$\begin{cases} x - 5 < 0 \\ x + 3 \geq 0 \end{cases} \cup \begin{cases} x - 5 \geq 0 \\ x - 3 > (x-5)^2 \end{cases}$$

$$\begin{cases} x < 5 \\ x \geq 3 \end{cases} \begin{cases} x \geq 5 \\ x - 3 > x^2 - 10x + 25 \end{cases}$$

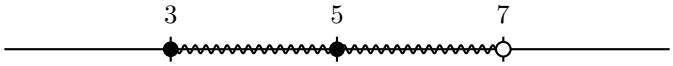
$$\begin{cases} x \geq 5 \\ x^2 - 11x + 28 < 0 \quad \text{II}^\circ \end{cases}$$

$$x^2 - 11x + 28 < 0 \quad (x-7)\,(x-4) < 0$$

$$\begin{cases} x < 5 \\ x \geq 3 \end{cases} \begin{cases} x \geq 5 \\ 4 < x < 7 \end{cases}$$



$3 \leq x < 5 \cup 5 \leq x < 7$



$\boxed{3 \leq x < 7}$